4.5 TABLE DESIGN

1.tbl_login

Primary key: **login_id**

Foreign key: **user_id** references table **tbl_user**

No:	Field name	Datatype (Size)	Key Constraints	Description of the field
1	login_id	Int(11)	Primary key, Auto_increment	Unique identifier for login.
2	email	Varchar(255)	Unique, Not Null	User's registered email address for login authentication.
3	password	Varchar(255)	Not Null	Securely hashed password string (bcrypt).
4	status	Enum	Not Null	Used to identify the user is Active, Inactive, or Suspended.
5	user_id	Int(11)	Foreign Key	References id of tbl_user.
6	role	Enum	Not Null	Used to identify the user is an admin, doctor, nurse, or patient.

2.tbl_user

Primary key: **user_id**

Foreign key: Nil

No:	Field name	Datatype (Size)	Key Constraints	Description of the field
1	user_id	Int(11)	Primary Key, Auto_increment	Unique Identifier for users.
2	name	Varchar(255)	Not Null	User's full name.
3	phone_no	Varchar(15)	Not Null	User's primary contact phone number.
4	created_at	Timestamp	Default Current_Timestamp	Timestamp when the user account was registered.
5	updated_at	Timestamp	Default Current_Timestamp	Timestamp when the user account was last updated.

3.tbl_role

Primary key: **role_id**

Foreign key: Nil

No:	Field name	Datatype (Size)	Key Constraints	Description of the field
1	role_id	Int(11)	Primary Key, Auto_increment	Unique identifier for system access roles.
2	role_name	Varchar(50)	Unique, Not Null	Role classification (admin, doctor, nurse, patient).
3	description	Varchar(255)	Null	Detailed summary of role capabilities and permissions.

4.tbl_gender

Primary key: **gender_id**

Foreign key: Nil

No:	Field name	Datatype (Size)	Key Constraints	Description of the field
1	gender_id	Int(11)	Primary Key, Auto_increment	Unique identifier for standardized gender.
2	gender_name	Varchar(20)	Unique, Not Null	Gender classification (Male, Female, Other, Prefer not to say).

5.tbl_blood_group

Primary key: **blood_group_id**

Foreign key: Nil

No:	Field name	Datatype (Size)	Key Constraints	Description of the field
1	blood_group_id	Int(11)	Primary Key, Auto_increment	Unique identifier for blood group type.
2	blood_group_name	Varchar(10)	Unique, Not Null	Standard clinical blood group label (A+, A-, B+, B-, AB+, AB-, O+, O-).

6.tbl_department

Primary key: **department_id**

Foreign key: Nil

No:	Field name	Datatype (Size)	Key Constraints	Description of the field
1	department_id	Int(11)	Primary Key, Auto_increment	Unique identifier for hospital department.
2	department_name	Varchar(150)	Not Null	Department title (Cardiology, Neurology, Pediatrics, Orthopedics, ICU).
3	description	Text	Null	Summary of medical services and clinical focus of the department.

7.tbl_specialization

Primary key: **specialization_id**

Foreign key: Nil

No:	Field name	Datatype (Size)	Key Constraints	Description of the field
1	specialization_id	Int(11)	Primary Key, Auto_increment	Unique identifier for medical specialization.
2	specialization_name	Varchar(150)	Unique, Not Null	Name of clinical specialty (General Medicine, Cardiology, Neurology, etc.).

8.tbl_doctor

Primary key: **doctor_id**

Foreign key: **user_id** references table **tbl_user**, **specialization_id** references table **tbl_specialization**, **department_id** references table **tbl_department**

No:	Field name	Datatype (Size)	Key Constraints	Description of the field
1	doctor_id	Int(11)	Primary Key, Auto_increment	Unique identifier for doctor profile.
2	user_id	Int(11)	Foreign Key, Unique, Not Null	References user_id of tbl_user.
3	first_name	Varchar(100)	Not Null	Doctor's first name.
4	last_name	Varchar(100)	Not Null	Doctor's last name or surname.
5	specialization_id	Int(11)	Foreign Key, Not Null	References specialization_id of tbl_specialization.
6	department_id	Int(11)	Foreign Key, Not Null	References department_id of tbl_department.
7	license_number	Varchar(100)	Unique, Not Null	Official state/national medical practitioner council registration number.
8	phone	Varchar(20)	Not Null	Doctor's direct professional telephone number.
9	years_of_experience	Int(11)	Not Null	Total number of years practicing clinical medicine.
10	created_at	Timestamp	Default Current_Timestamp	Record registration timestamp.

9.tbl_nurse

Primary key: **nurse_id**

Foreign key: **user_id** references table **tbl_user**, **department_id** references table **tbl_department**

No:	Field name	Datatype (Size)	Key Constraints	Description of the field
1	nurse_id	Int(11)	Primary Key, Auto_increment	Unique identifier for nurse record.
2	user_id	Int(11)	Foreign Key, Unique, Not Null	References user_id of tbl_user.
3	first_name	Varchar(100)	Not Null	Nurse's first name.
4	last_name	Varchar(100)	Not Null	Nurse's last name.
5	department_id	Int(11)	Foreign Key, Not Null	References department_id of tbl_department.
6	license_number	Varchar(100)	Unique, Not Null	Nursing council official license number.
7	nurse_code	Varchar(50)	Unique, Not Null	Internal hospital employee badge identification code.
8	registration_number	Varchar(100)	Unique, Not Null	Registered Nurse (RN) statutory credential number.
9	assigned_ward	Varchar(100)	Null	Hospital inpatient ward or nursing station assigned for duties.
10	phone	Varchar(20)	Not Null	Nurse's contact telephone number.

10.tbl_admin

Primary key: **admin_id**

Foreign key: **user_id** references table **tbl_user**

No:	Field name	Datatype (Size)	Key Constraints	Description of the field
1	admin_id	Int(11)	Primary Key, Auto_increment	Unique identifier for administrator record.
2	user_id	Int(11)	Foreign Key, Unique, Not Null	References user_id of tbl_user.
3	first_name	Varchar(100)	Not Null	Administrator's first name.
4	last_name	Varchar(100)	Not Null	Administrator's last name.
5	phone	Varchar(20)	Not Null	Administrator's direct telephone contact number.
6	created_at	Timestamp	Default Current_Timestamp	Administrative account registration timestamp.

11.tbl_patient

Primary key: **patient_id**

Foreign key: **user_id** references table **tbl_user**, **gender_id** references table **tbl_gender**, **blood_group_id** references table **tbl_blood_group**

No:	Field name	Datatype (Size)	Key Constraints	Description of the field
1	patient_id	Int(11)	Primary Key, Auto_increment	Unique identifier for patient health profile.
2	user_id	Int(11)	Foreign Key, Unique, Not Null	References user_id of tbl_user.
3	first_name	Varchar(100)	Not Null	Patient's given first name.
4	last_name	Varchar(100)	Not Null	Patient's family name or surname.
5	date_of_birth	Date	Not Null	Legal calendar birth date used for age calculations.
6	gender_id	Int(11)	Foreign Key, Not Null	References gender_id of tbl_gender.
7	blood_group_id	Int(11)	Foreign Key, Not Null	References blood_group_id of tbl_blood_group.
8	phone	Varchar(20)	Not Null	Patient contact telephone number.
9	address	Text	Not Null	Residential street address.
10	city	Varchar(100)	Not Null	City of residence.
11	state	Varchar(100)	Not Null	State or administrative district.
12	emergency_contact_name	Varchar(200)	Not Null	Full name of appointed emergency contact person.
13	emergency_contact_phone	Varchar(20)	Not Null	Phone number of emergency contact.
14	admission_status	Varchar(50)	Not Null	Current hospital care status (Outpatient, Inpatient, Discharged).

12.tbl_doctor_availability

Primary key: **availability_id**

Foreign key: **doctor_id** references table **tbl_doctor**

No:	Field name	Datatype (Size)	Key Constraints	Description of the field
1	availability_id	Int(11)	Primary Key, Auto_increment	Unique identifier for doctor schedule entry.
2	doctor_id	Int(11)	Foreign Key, Not Null	References doctor_id of tbl_doctor.
3	date	Date	Not Null	Calendar date for the clinical duty shift.
4	status	Varchar(50)	Not Null	Duty roster state (AVAILABLE, ON_LEAVE, BUSY).
5	start_time	Varchar(10)	Not Null	Duty start time (e.g., 09:00).
6	end_time	Varchar(10)	Not Null	Duty end time (e.g., 17:00).
7	reason	Varchar(255)	Null	Stated justification if doctor is on leave or unavailable.

13.tbl_appointment_status

Primary key: **status_id**

Foreign key: Nil

No:	Field name	Datatype (Size)	Key Constraints	Description of the field
1	status_id	Int(11)	Primary Key, Auto_increment	Unique identifier for appointment status.
2	name	Varchar(50)	Unique, Not Null	Appointment state designation (scheduled, completed, cancelled, no_show).

14.tbl_appointment

Primary key: **appointment_id**

Foreign key: **patient_id** references table **tbl_patient**, **doctor_id** references table **tbl_doctor**, **status_id** references table **tbl_appointment_status**

No:	Field name	Datatype (Size)	Key Constraints	Description of the field
1	appointment_id	Int(11)	Primary Key, Auto_increment	Unique identifier for appointment booking.
2	patient_id	Int(11)	Foreign Key, Not Null	References patient_id of tbl_patient.
3	doctor_id	Int(11)	Foreign Key, Not Null	References doctor_id of tbl_doctor.
4	appointment_date	Date	Not Null	Calendar date of scheduled clinical visit.
5	appointment_time	Time	Not Null	Time slot for clinical consultation.
6	status_id	Int(11)	Foreign Key, Not Null	References status_id of tbl_appointment_status.
7	reason	Text	Not Null	Primary medical complaint or reason for booking.
8	notes	Text	Null	Physician clinical findings, observations, or preparatory notes.

15.tbl_medicine

Primary key: **medicine_id**

Foreign key: Nil

No:	Field name	Datatype (Size)	Key Constraints	Description of the field
1	medicine_id	Int(11)	Primary Key, Auto_increment	Unique identifier for pharmaceutical medicine.
2	name	Varchar(255)	Not Null	Commercial brand name of the drug.
3	generic_name	Varchar(255)	Null	Standard generic pharmaceutical salt composition name.
4	category	Varchar(100)	Not Null	Drug classification (Antibiotic, Analgesic, Antipyretic, Antihypertensive).
5	unit	Varchar(50)	Not Null	Dispensation package unit (Tablet, Capsule, Syrup, Injection, Cream).
6	description	Text	Null	Pharmacological overview, storage guidelines, and contraindications.

16.tbl_prescription

Primary key: **prescription_id**

Foreign key: **patient_id** references table **tbl_patient**, **doctor_id** references table **tbl_doctor**, **appointment_id** references table **tbl_appointment**

No:	Field name	Datatype (Size)	Key Constraints	Description of the field
1	prescription_id	Int(11)	Primary Key, Auto_increment	Unique identifier for medical prescription header.
2	patient_id	Int(11)	Foreign Key, Not Null	References patient_id of tbl_patient.
3	doctor_id	Int(11)	Foreign Key, Not Null	References doctor_id of tbl_doctor.
4	appointment_id	Int(11)	Foreign Key, Null	References appointment_id of tbl_appointment.
5	diagnosis	Text	Not Null	Clinical diagnostic conclusion determined by physician.
6	notes	Text	Null	Physician remarks, dietary suggestions, and recovery advice.
7	prescribed_date	Date	Not Null	Calendar date prescription was formally issued.
8	valid_until	Date	Null	Prescription validity or pharmacy dispensing expiration date.

17.tbl_prescription_item

Primary key: **item_id**

Foreign key: **prescription_id** references table **tbl_prescription**, **medicine_id** references table **tbl_medicine**

No:	Field name	Datatype (Size)	Key Constraints	Description of the field
1	item_id	Int(11)	Primary Key, Auto_increment	Unique identifier for prescription line item.
2	prescription_id	Int(11)	Foreign Key, Not Null	References prescription_id of tbl_prescription.
3	medicine_id	Int(11)	Foreign Key, Not Null	References medicine_id of tbl_medicine.
4	dosage	Varchar(100)	Not Null	Specific strength and dosage quantity (e.g., 500mg, 1 tablet).
5	frequency	Varchar(100)	Not Null	Schedule frequency (e.g., Once daily, 1-0-1, Every 8 hours).
6	duration_days	Int(11)	Not Null	Number of days the drug should be consumed.
7	instructions	Text	Null	Consumption instructions (e.g., Take after food with warm water).

18.tbl_record_type

Primary key: **record_type_id**

Foreign key: Nil

No:	Field name	Datatype (Size)	Key Constraints	Description of the field
1	record_type_id	Int(11)	Primary Key, Auto_increment	Unique identifier for clinical record type.
2	name	Varchar(100)	Unique, Not Null	Record classification (consultation, lab_result, surgery, radiology, vaccination).

19.tbl_medical_record

Primary key: **record_id**

Foreign key: **patient_id** references table **tbl_patient**, **doctor_id** references table **tbl_doctor**, **appointment_id** references table **tbl_appointment**, **record_type_id** references table **tbl_record_type**

No:	Field name	Datatype (Size)	Key Constraints	Description of the field
1	record_id	Int(11)	Primary Key, Auto_increment	Unique identifier for electronic health record.
2	patient_id	Int(11)	Foreign Key, Not Null	References patient_id of tbl_patient.
3	doctor_id	Int(11)	Foreign Key, Not Null	References doctor_id of tbl_doctor.
4	appointment_id	Int(11)	Foreign Key, Null	References appointment_id of tbl_appointment.
5	record_type_id	Int(11)	Foreign Key, Not Null	References record_type_id of tbl_record_type.
6	title	Varchar(255)	Not Null	Short descriptive title of the medical examination.
7	description	Text	Not Null	Comprehensive clinical findings, exam notes, and evaluations.
8	record_date	Date	Not Null	Calendar date the examination or procedure took place.

20.tbl_document_type

Primary key: **document_type_id**

Foreign key: Nil

No:	Field name	Datatype (Size)	Key Constraints	Description of the field
1	document_type_id	Int(11)	Primary Key, Auto_increment	Unique identifier for document format type.
2	name	Varchar(100)	Unique, Not Null	Document classification (X-Ray, MRI, Lab Report, ECG, Prescription_PDF).

21.tbl_medical_document

Primary key: **document_id**

Foreign key: **patient_id** references table **tbl_patient**, **record_id** references table **tbl_medical_record**, **document_type_id** references table **tbl_document_type**, **uploaded_by_id** references table **tbl_user**

No:	Field name	Datatype (Size)	Key Constraints	Description of the field
1	document_id	Int(11)	Primary Key, Auto_increment	Unique identifier for uploaded medical document.
2	patient_id	Int(11)	Foreign Key, Not Null	References patient_id of tbl_patient.
3	record_id	Int(11)	Foreign Key, Null	References record_id of tbl_medical_record.
4	document_type_id	Int(11)	Foreign Key, Not Null	References document_type_id of tbl_document_type.
5	file_name	Varchar(255)	Not Null	Original filename of the stored diagnostic document.
6	file_path	Text	Not Null	Secure Cloudinary URL or local file storage system path.
7	file_size_kb	Int(11)	Not Null	File storage footprint size measured in kilobytes.
8	uploaded_by_id	Int(11)	Foreign Key, Not Null	References user_id of tbl_user who performed file upload.
9	uploaded_at	Timestamp	Default Current_Timestamp	Timestamp when document was stored in system.

22.tbl_ward

Primary key: **ward_id**

Foreign key: Nil

No:	Field name	Datatype (Size)	Key Constraints	Description of the field
1	ward_id	Int(11)	Primary Key, Auto_increment	Unique identifier for hospital inpatient ward.
2	ward_number	Varchar(50)	Unique, Not Null	Distinct clinical ward code (e.g., WARD-GEN-A, ICU-01).
3	name	Varchar(150)	Not Null	Designation name of the ward (e.g., General Ward, Post-Op Care).
4	department	Varchar(150)	Null	Medical department supervising the ward operations.
5	total_beds	Int(11)	Not Null	Total bed installation capacity in this ward.
6	is_active	Boolean	Default True	Operational availability status of the ward.

23.tbl_bed

Primary key: **bed_id**

Foreign key: **ward_id** references table **tbl_ward**

No:	Field name	Datatype (Size)	Key Constraints	Description of the field
1	bed_id	Int(11)	Primary Key, Auto_increment	Unique identifier for hospital bed.
2	ward_id	Int(11)	Foreign Key, Not Null	References ward_id of tbl_ward.
3	bed_number	Varchar(50)	Not Null	Physical bed identifier code (e.g., B-101, ICU-04).
4	status	Varchar(50)	Not Null	Current occupancy state (AVAILABLE, OCCUPIED, MAINTENANCE).

24.tbl_inpatient_admission

Primary key: **admission_id**

Foreign key: **patient_id** references table **tbl_patient**, **admitting_doctor_id** references table **tbl_doctor**, **ward_id** references table **tbl_ward**, **bed_id** references table **tbl_bed**

No:	Field name	Datatype (Size)	Key Constraints	Description of the field
1	admission_id	Int(11)	Primary Key, Auto_increment	Unique identifier for inpatient admission record.
2	admission_number	Varchar(50)	Unique, Not Null	Formal admission case identification number.
3	patient_id	Int(11)	Foreign Key, Not Null	References patient_id of tbl_patient.
4	admitting_doctor_id	Int(11)	Foreign Key, Not Null	References doctor_id of tbl_doctor who admitted patient.
5	ward_id	Int(11)	Foreign Key, Not Null	References ward_id of tbl_ward.
6	bed_id	Int(11)	Foreign Key, Not Null	References bed_id of assigned tbl_bed.
7	admission_date_time	Timestamp	Default Current_Timestamp	Exact date and time when patient was admitted.
8	admission_diagnosis	Varchar(255)	Not Null	Provisional medical diagnosis justifying inpatient stay.
9	chief_complaint	Text	Not Null	Clinical symptoms presented upon emergency/ward arrival.
10	admission_status	Varchar(50)	Not Null	Inpatient status (ADMITTED, TRANSFERRED, DISCHARGED).
11	acuity_level	Varchar(50)	Not Null	Clinical urgency level (CRITICAL, MODERATE, STABLE).
12	discharge_date	Timestamp	Null	Timestamp of formal medical discharge from hospital.

25.tbl_bed_transfer_history

Primary key: **transfer_id**

Foreign key: **admission_id** references table **tbl_inpatient_admission**, **from_ward_id** references table **tbl_ward**, **from_bed_id** references table **tbl_bed**, **to_ward_id** references table **tbl_ward**, **to_bed_id** references table **tbl_bed**, **transferred_by_id** references table **tbl_user**

No:	Field name	Datatype (Size)	Key Constraints	Description of the field
1	transfer_id	Int(11)	Primary Key, Auto_increment	Unique identifier for bed transfer audit trail.
2	admission_id	Int(11)	Foreign Key, Not Null	References admission_id of tbl_inpatient_admission.
3	from_ward_id	Int(11)	Foreign Key, Not Null	References ward_id of origin ward.
4	from_bed_id	Int(11)	Foreign Key, Not Null	References bed_id of origin bed.
5	to_ward_id	Int(11)	Foreign Key, Not Null	References ward_id of target ward.
6	to_bed_id	Int(11)	Foreign Key, Not Null	References bed_id of target bed.
7	transfer_reason	Varchar(255)	Null	Clinical or administrative justification for patient transfer.
8	transferred_by_id	Int(11)	Foreign Key, Not Null	References user_id of authorized staff who initiated move.
9	transferred_at	Timestamp	Default Current_Timestamp	Timestamp when patient bed transfer was executed.

26.tbl_patient_observation

Primary key: **observation_id**

Foreign key: **patient_id** references table **tbl_patient**, **nurse_id** references table **tbl_nurse**

No:	Field name	Datatype (Size)	Key Constraints	Description of the field
1	observation_id	Int(11)	Primary Key, Auto_increment	Unique identifier for vital signs observation record.
2	patient_id	Int(11)	Foreign Key, Not Null	References patient_id of tbl_patient.
3	nurse_id	Int(11)	Foreign Key, Not Null	References nurse_id of tbl_nurse who recorded vitals.
4	observation_date	Date	Not Null	Calendar date observation was conducted.
5	observation_time	Time	Not Null	Time of vitals recording.
6	temperature	Decimal(5,2)	Not Null	Body temperature measurement in Fahrenheit / Celsius.
7	pulse_rate	Int(11)	Not Null	Heart pulse rate in beats per minute (BPM).
8	respiratory_rate	Int(11)	Not Null	Respiratory rate in breaths per minute.
9	systolic_bp	Int(11)	Not Null	Systolic blood pressure in mmHg.
10	diastolic_bp	Int(11)	Not Null	Diastolic blood pressure in mmHg.
11	spo2	Decimal(5,2)	Not Null	Peripheral arterial oxygen saturation (%).
12	blood_glucose	Decimal(7,2)	Null	Blood sugar concentration in mg/dL.
13	weight	Decimal(6,2)	Null	Patient body weight in kilograms.
14	pain_score	Int(11)	Null	Visual / numeric pain rating score (0 to 10).
15	consciousness_level	Varchar(50)	Not Null	Consciousness status (Alert, Confused, Drowsy, Unresponsive).
16	general_observation	Text	Null	Clinical nursing evaluation notes on patient condition.

27.tbl_discharge_summary

Primary key: **summary_id**

Foreign key: **patient_id** references table **tbl_patient**, **doctor_id** references table **tbl_doctor**

No:	Field name	Datatype (Size)	Key Constraints	Description of the field
1	summary_id	Int(11)	Primary Key, Auto_increment	Unique identifier for patient discharge summary.
2	patient_id	Int(11)	Foreign Key, Not Null	References patient_id of tbl_patient.
3	doctor_id	Int(11)	Foreign Key, Not Null	References doctor_id of attending physician.
4	admission_date	Date	Not Null	Date of inpatient hospital admission.
5	discharge_date	Date	Not Null	Date patient was certified for discharge.
6	admission_diagnosis	Varchar(255)	Null	Provisional diagnosis documented upon admission.
7	discharge_diagnosis	Varchar(255)	Not Null	Final confirmed diagnosis at completion of stay.
8	clinical_course	Text	Not Null	Narrative summary of treatments and inpatient clinical course.
9	condition_at_discharge	Varchar(100)	Not Null	Physical status at discharge (Improved, Stable, Cured).
10	follow_up_instructions	Text	Null	Post-discharge rehabilitation and medical instructions.
11	follow_up_date	Date	Null	Recommended date for clinical follow-up review.
12	summary_status	Varchar(50)	Not Null	Discharge summary document status (DRAFT, FINALIZED).

28.tbl_nursing_patient_summary

Primary key: **summary_id**

Foreign key: **patient_id** references table **tbl_patient**, **nurse_id** references table **tbl_nurse**

No:	Field name	Datatype (Size)	Key Constraints	Description of the field
1	summary_id	Int(11)	Primary Key, Auto_increment	Unique identifier for nursing shift handover report.
2	patient_id	Int(11)	Foreign Key, Not Null	References patient_id of tbl_patient.
3	nurse_id	Int(11)	Foreign Key, Not Null	References nurse_id of authoring nurse.
4	summary_date	Date	Not Null	Calendar date of nursing shift handover.
5	patient_current_condition	Varchar(255)	Not Null	Physical condition and mobility assessment.
6	level_of_consciousness	Varchar(50)	Null	Mental alertness and neurological response status.
7	vital_signs_summary	Text	Null	Overview of vital signs observed throughout the shift.
8	nursing_care_provided	Text	Null	Care interventions, dressings, and medication delivery administered.
9	status	Varchar(50)	Not Null	Handover report state (DRAFT, FINALIZED).

29.tbl_reminder_status

Primary key: **status_id**

Foreign key: Nil

No:	Field name	Datatype (Size)	Key Constraints	Description of the field
1	status_id	Int(11)	Primary Key, Auto_increment	Unique identifier for medication reminder status.
2	name	Varchar(50)	Unique, Not Null	Reminder status (ACTIVE, TAKEN, MISSED, CANCELLED).

30.tbl_medicine_reminder

Primary key: **reminder_id**

Foreign key: **patient_id** references table **tbl_patient**, **medicine_id** references table **tbl_medicine**, **status_id** references table **tbl_reminder_status**

No:	Field name	Datatype (Size)	Key Constraints	Description of the field
1	reminder_id	Int(11)	Primary Key, Auto_increment	Unique identifier for medication reminder alert.
2	patient_id	Int(11)	Foreign Key, Not Null	References patient_id of tbl_patient.
3	medicine_id	Int(11)	Foreign Key, Not Null	References medicine_id of tbl_medicine.
4	reminder_time	Time	Not Null	Daily scheduled alert time for medication intake.
5	frequency	Varchar(100)	Not Null	Administration schedule (Once daily, Twice daily, Every 8 hours).
6	start_date	Date	Not Null	Commencement date for reminder alerts.
7	end_date	Date	Null	Concluding date for medication schedule.
8	status_id	Int(11)	Foreign Key, Not Null	References status_id of tbl_reminder_status.

31.tbl_notification_type

Primary key: **notification_type_id**

Foreign key: Nil

No:	Field name	Datatype (Size)	Key Constraints	Description of the field
1	notification_type_id	Int(11)	Primary Key, Auto_increment	Unique identifier for notification category.
2	name	Varchar(100)	Unique, Not Null	Category classification (APPOINTMENT, PRESCRIPTION, REMINDER, SYSTEM).

32.tbl_notification

Primary key: **notification_id**

Foreign key: **user_id** references table **tbl_user**, **notification_type_id** references table **tbl_notification_type**

No:	Field name	Datatype (Size)	Key Constraints	Description of the field
1	notification_id	Int(11)	Primary Key, Auto_increment	Unique identifier for system user notification.
2	user_id	Int(11)	Foreign Key, Not Null	References user_id of recipient tbl_user.
3	notification_type_id	Int(11)	Foreign Key, Not Null	References notification_type_id of tbl_notification_type.
4	title	Varchar(255)	Not Null	Subject title of the notification message.
5	message	Text	Not Null	Full textual body of the alert notification.
6	is_read	Boolean	Default False	Read acknowledgment status flag.
7	created_at	Timestamp	Default Current_Timestamp	Timestamp when notification was created.

33.tbl_hospital_procedure

Primary key: **procedure_id**

Foreign key: **department_id** references table **tbl_department**

No:	Field name	Datatype (Size)	Key Constraints	Description of the field
1	procedure_id	Int(11)	Primary Key, Auto_increment	Unique identifier for hospital procedure / protocol.
2	department_id	Int(11)	Foreign Key, Null	References department_id of tbl_department.
3	title	Varchar(255)	Not Null	Official clinical procedure protocol title.
4	procedure_code	Varchar(50)	Unique, Not Null	Hospital protocol reference code (e.g., SOP-CARD-01).
5	category	Varchar(100)	Not Null	Protocol category (Emergency, Infection Control, Surgical).
6	version	Varchar(50)	Not Null	Protocol version number (e.g., 1.0, 2.0).
7	status	Varchar(50)	Not Null	Review status (PUBLISHED, DRAFT, UNDER_REVIEW).
8	effective_date	Date	Not Null	Effective implementation date for protocol.

34.tbl_action_type

Primary key: **action_type_id**

Foreign key: Nil

No:	Field name	Datatype (Size)	Key Constraints	Description of the field
1	action_type_id	Int(11)	Primary Key, Auto_increment	Unique identifier for audit action type.
2	name	Varchar(50)	Unique, Not Null	Action classification (CREATE, READ, UPDATE, DELETE, LOGIN, LOGOUT).

35.tbl_audit_log

Primary key: **audit_id**

Foreign key: **user_id** references table **tbl_user**, **action_type_id** references table **tbl_action_type**

No:	Field name	Datatype (Size)	Key Constraints	Description of the field
1	audit_id	Int(11)	Primary Key, Auto_increment	Unique identifier for system audit log.
2	user_id	Int(11)	Foreign Key, Null	References user_id of tbl_user who performed operation.
3	action_type_id	Int(11)	Foreign Key, Not Null	References action_type_id of tbl_action_type.
4	table_name	Varchar(100)	Not Null	Database entity table affected by the action.
5	record_id	Int(11)	Null	Primary key ID altered in the target table.
6	ip_address	Varchar(45)	Null	Network client IP address of requester.
7	created_at	Timestamp	Default Current_Timestamp	Timestamp when audit event occurred.